Individual Lab (Case study 2)

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Cohort; A

QUESTION 7.24

```
create table department(
deptNo int primary key,
deptName varchar(20) not null,
mgrEmpNo varchar(3)
);

create table employee(
empNo varchar(3) primary key,
fName varchar(30) not null,
lName varchar(30) not null,
address varchar(50),
DOB date,
sex char(1) not null,
position varchar(20) not null,
deptNo int,
foreign key(deptNo) references department(deptNo),
check (sex in('F', 'M')),
check (position in('Manager', 'Team Leader','Analyst','Software Developer'))
);
```

```
-- updating department due to circular foreign key
alter table department
add foreign key(mgrEmpNo) references employee(empNo);

create table project(
projNo int primary key,
projName varchar(50) not  null,
deptNo int,
foreign key(deptNo) references department(deptNo)
);


create table worksOn(
empNo varchar(3),
projNo int,
dateWorked date,
hoursWorked int,
primary key (empNo, projNo, dateWorked),
foreign key (empNo) references employee(empNo),
foreign key(projNo) references project(projNo),
check (hoursWorked between 0 and 40)
);

insert into department values (1, 'it', null);
insert into department values(2, 'finance', null);
insert into department values (3, 'hr', null);
insert into department values (4, 'marketing', null);
insert into department values (5, 'sales', null);
```

```
insert into department values (6, 'logistics', null);
insert into department values (7, 'legal', null);
insert into department values (8, 'accounting', null);
insert into department values (9, 'operations', null);
insert into department values (10, 'security', null);
insert into department values (11, 'research', null);
insert into department values (12, 'support', null);
```

output:

```
MariaDB [employee_management]> select * from department;
+--------+------------+----------+
| deptNo | deptName   | mgrEmpNo |
+--------+------------+----------+
|      1 | it         | E1       |
|      2 | finance    | NULL     |
|      3 | hr         | NULL     |
|      4 | marketing  | NULL     |
|      5 | sales      | E5       |
|      6 | logistics  | E6       |
|      7 | legal      | NULL     |
|      8 | accounting | NULL     |
|      9 | operations | NULL     |
|     10 | security   | NULL     |
|     11 | research   | E11      |
|     12 | support    | NULL     |
+--------+------------+----------+
12 rows in set (0.002 sec)
```

```
-- triger for constriant :There is only 1 manager for each department
DELIMITER //
create trigger one_manager_per_department
before insert on employee
for each row
begin
   if new.position = 'Manager' then
      if exists (
        select *
         from employee
         where deptNo = new.deptNo
          and position = 'Manager'
      ) then
         signal sqlstate '45000'
         set message_text = 'Only one manager per department';
      end if;
   end if ;
end//

DELIMITER ;

insert into employee values ('E1', 'john', 'doe', '123 main st', '1985-03-15', 'M', 'Manager', 1);
insert into employee values ('E2', 'jane', 'smith', '456 oak ave', '1990-07-22', 'F', 'Analyst', 2);
insert into employee values ('E3', 'bob', 'brown', '789 pine rd', '1988-11-10', 'M', 'Software Developer', 3);
```

insert into employee values ('E4', 'alice', 'white', '321 elm st', '1992-05-18', 'F', 'Team Leader', 4);

insert into employee values ('E5', 'mike', 'jones', '654 cedar ln', '1987-09-30', 'M', 'Manager', 5);

insert into employee values ('E6', 'sara', 'davis', '987 birch blvd', '1993-02-14', 'F', 'Manager', 6);

insert into employee values ('E7', 'tom', 'wilson', '111 maple dr', '1986-06-25', 'M', 'Software Developer', 7);

insert into employee values ('E8', 'linda', 'moore', '222 walnut ave', '1991-08-19', 'F', 'Analyst', 8);

insert into employee values ('E9', 'james', 'taylor', '333 spruce st', '1984-12-05', 'M', 'Team Leader', 9);

insert into employee values ('E10', 'emma', 'anderson', '444 pine ave', '1995-04-30', 'F', 'Software Developer', 10);

insert into employee values ('E11', 'paul', 'thomas', '555 oak blvd', '1983-07-11', 'M', 'Manager', 11);

insert into employee values ('E12', 'lucy', 'jackson', '666 elm rd', '1994-01-22', 'F', 'Analyst', 12);

output:

MariaDB [employee_management]> select * from employee;

+-------+-------+----------+----------------+------------+-----+--------------------+--------+

| empNo | fName | lName | address | DOB | sex | position | deptNo |

+-------+-------+----------+----------------+------------+-----+--------------------+--------+

| E1 | john | doe | 123 main st | 1985-03-15 | M | Manager | 1 |

| E10 | emma | anderson | 444 pine ave | 1995-04-30 | F | Software Developer | 10 |

| E11 | paul | thomas | 555 oak blvd | 1983-07-11 | M | Manager | 11 |

| E12 | lucy | jackson | 666 elm rd | 1994-01-22 | F | Analyst | 12 |

| E2 | jane | smith | 456 oak ave | 1990-07-22 | F | Analyst | 2 |

| E3 | bob | brown | 789 pine rd | 1988-11-10 | M | Software Developer | 3 |

| E4 | alice | white | 321 elm st | 1992-05-18 | F | Team Leader | 4 |

| E5 | mike | jones | 654 cedar ln | 1987-09-30 | M | Manager | 5 |

| E6 | sara | davis | 987 birch blvd | 1993-02-14 | F | Manager | 6 |

| E7 | tom | wilson | 111 maple dr | 1986-06-25 | M | Software Developer | 7 |

| E8 | linda | moore | 222 walnut ave | 1991-08-19 | F | Analyst | 8 |

| E9 | james | taylor | 333 spruce st | 1984-12-05 | M | Team Leader | 9 |

+-------+-------+----------+----------------+------------+-----+--------------------+--------+

12 rows in set (0.001 sec)

```
-- trigger to check that mgrEmpNo are allocated only to employees which are managers
DELIMITER //
create trigger check_manager_assignment
before update on department
for each row
begin
if new.mgrEmpNo is not null then
if not exists (
select *
from employee
where empNo = new.mgrEmpNo
and position = 'Manager') then
signal sqlstate '45000'
set message_text= 'Only employees with position Manager can be assigned as
department manager';
end if;
end if;
end//
DELIMITER ;
```

```
update department set mgrempno = 'E1' where deptno = 1;
update department set mgrempno = 'E5' where deptno = 5;
update department set mgrempno = 'E6' where deptno = 6;
update department set mgrempno = 'E11' where deptno = 11;
```

output:

```
MariaDB [employee_management]> select * from department;
+--------+------------+----------+
| deptNo | deptName   | mgrEmpNo |
+--------+------------+----------+
|      1 | it         | E1       |
|      2 | finance    | NULL     |
|      3 | hr         | NULL     |
|      4 | marketing  | NULL     |
|      5 | sales      | E5       |
|      6 | logistics  | E6       |
|      7 | legal      | NULL     |
|      8 | accounting | NULL     |
|      9 | operations | NULL     |
|     10 | security   | NULL     |
|     11 | research   | E11      |
|     12 | support    | NULL     |
+--------+------------+----------+
12 rows in set (0.002 sec)
```

```
insert into project values (1, 'sccs', 1);
insert into project values (2, 'payroll system', 2);
```

```
insert into project values (3, 'hr portal', 3);
insert into project values (4, 'marketing app', 4);
insert into project values (5, 'sales tracker', 5);
insert into project values (6, 'logistics tool', 6);
insert into project values (7, 'legal database', 7);
insert into project values (8, 'accounting system', 8);
insert into project values (9, 'operations hub', 9);
insert into project values (10, 'security monitor', 10);
insert into project values (11, 'research portal', 11);
insert into project values (12, 'support desk', 12);
```

output:

```
MariaDB [employee_management]> select * from project;
+--------+------------------+--------+
| projNo | projName         | deptNo |
+--------+------------------+--------+
|      1 | sccs             |      1 |
|      2 | payroll system   |      2 |
|      3 | hr portal        |      3 |
|      4 | marketing app    |      4 |
|      5 | sales tracker    |      5 |
|      6 | logistics tool   |      6 |
|      7 | legal database   |      7 |
|      8 | accounting system |      8 |
|      9 | operations hub   |      9 |
|     10 | security monitor |     10 |
|     11 | research portal  |     11 |
|     12 | support desk     |     12 |
```

+--------+-------------------+--------+

insert into workson values ('E1', 1, '2024-01-10', 8);

insert into workson values ('E2', 2, '2024-01-11', 6);

insert into workson values ('E12', 1, '2024-01-12', 7);

insert into workson values ('E3', 1, '2024-01-13', 5);

insert into workson values ('E4', 3, '2024-01-14', 4);

insert into workson values ('E5', 1, '2024-01-15', 4);

insert into workson values ('E10', 2, '2024-01-16', 6);

insert into workson values ('E6', 4, '2024-01-17', 7);

insert into workson values ('E7', 5, '2024-01-18', 5);

insert into workson values ('E8', 6, '2024-01-19', 8);

insert into workson values ('E12', 7, '2024-01-20', 6);

insert into workson values ('E11', 8, '2024-01-21', 4);

output:

MariaDB [employee_management]> select * from workson;

+-------+--------+------------+-------------+

| empNo | projNo | dateWorked | hoursWorked |

+-------+--------+------------+-------------+

| E1 | 1 | 2024-01-10 | 8 |

| E10 | 2 | 2024-01-16 | 6 |

| E11 | 8 | 2024-01-21 | 4 |

| E12 | 1 | 2024-01-12 | 7 |

| E12 | 7 | 2024-01-20 | 6 |

| E2 | 2 | 2024-01-11 | 6 |

| E3 | 1 | 2024-01-13 | 5 |

| E4 | 3 | 2024-01-14 | 4 |

| E5 | 1 | 2024-01-15 | 4 |

| E6 | 4 | 2024-01-17 | 7 |

| E7 | 5 | 2024-01-18 | 5 |

| E8 | 6 | 2024-01-19 | 8 |

+-------+--------+------------+-------------+

12 rows in set (0.001 sec)

question 7.25

```
-- view for projects managed by female managers
create view female_managers_project as
select w.projNo, p.projname, e.fname, e.lname, e.sex, d.mgrEmpno
from department d
inner join employee e
on d.mgrEmpNo = e.empNo
inner join workson w
on e.empNo = w.empNo
inner join project p
on w.projNo = p.projNo
where e.sex = 'F';
```

output:

```
MariaDB [employee_management]> select * from female_managers_project
   -> order by projNo;
+--------+---------------+-------+-------+-----+----------+
| projNo | projname      | fname | lname | sex | mgrEmpno |
+--------+---------------+-------+-------+-----+----------+
```

```
|     4 | marketing app | sara  | davis | F   | E6      |
+--------+---------------+-------+-------+-----+-----------+
1 row in set (0.003 sec)
```

question 7.26

```
-- employee details view

create view emp_details as
select w.empNo, e.fName, e.lName, p.projName, w.hoursWorked
from employee e
inner join worksOn w
on e.empNo = w.empNo
inner join project p
on w.projNo = p.projNo;
```

output:

```
MariaDB [employee_management]> select * from emp_details;
+-------+-------+----------+------------------+-------------+
| empNo | fName | lName    | projName         | hoursWorked |
+-------+-------+----------+------------------+-------------+
| E1    | john  | doe      | sccs             |         8 |
| E10   | emma  | anderson | payroll system   |         6 |
| E11   | paul  | thomas   | accounting system |         4 |
| E12   | lucy  | jackson  | sccs             |         7 |
| E12   | lucy  | jackson  | legal database   |         6 |
| E2    | jane  | smith    | payroll system   |         6 |
| E3    | bob   | brown    | sccs             |         5 |
```

| E4 | alice | white | hr portal | 4 |

| E5 | mike | jones | sccs | 4 |

| E6 | sara | davis | marketing app | 7 |

| E7 | tom | wilson | sales tracker | 5 |

| E8 | linda | moore | logistics tool | 8 |

+-------+-------+----------+------------------+-------------+

12 rows in set (0.002 sec)

Question 7.27

```
create view empproject(empno, projno, totalhours)
as select w.empno, w.projno, sum(hoursworked)
from employee e, project p, workson w
where e.empno = w.empno and p.projno = w.projno
group by w.empno, w.projno;
```

output:

```
MariaDB [employee_management]> select * from empproject;
+-------+--------+------------+
| empno | projno | totalhours |
+-------+--------+------------+
| E1    |      1 |          8 |
| E10   |      2 |          6 |
| E11   |      8 |          4 |
| E12   |      1 |          7 |
| E12   |      7 |          6 |
| E2    |      2 |          6 |
| E3    |      1 |          5 |
```

```
| E4   |     3 |        4 |
| E5   |     1 |        4 |
| E6   |     4 |        7 |
| E7   |     5 |        5 |
| E8   |     6 |        8 |
+-------+--------+------------+
12 rows in set (0.004 sec)

MariaDB [employee_management]> select count(projNo)
   -> from empproject
   -> where empNo = 'E1';
+---------------+
| count(projNo) |
+---------------+
|           1 |
+---------------+
1 row in set (0.003 sec)
```

output:

```
MariaDB [employee_management]> select empNo, sum(totalhours) as totalHours
   -> from empproject
   -> group by empNo;
+-------+------------+
| empNo | totalHours |
+-------+------------+
| E1    |        8 |
| E10   |        6 |
```

| E11 | 4 |

| E12 | 13 |

| E2 | 6 |

| E3 | 5 |

| E4 | 4 |

| E5 | 4 |

| E6 | 7 |

| E7 | 5 |

| E8 | 8 |

+-------+------------+

11 rows in set (0.002 sec)

Question 7.28

Materialized View Maintenance of ExpensiveParts

A materialized view physically stores the results of a query unlike a regular view which is virtual. To maintain the ExpensiveParts materialized view, any changes made to the base table Part must be reflected in the stored results.

The view contains distinct part numbers where partCost > 1000. Since a part can have different prices under different contracts, maintenance depends on the type of change made to the Part table.

1. INSERT

- if new partCost > 1000, add partNo to view (no need to access base table)
- if new partCost <= 1000, do nothing (no need to access base table)

2. UPDATE

- if partCost updated above 1000, add partNo to view (no need to access base table)
- if partCost updated below 1000, MUST access base table to check if partNo still exists under another contract with partCost > 1000

3. DELETE

- MUST access base table to check if partNo still exists under another contract with partCost > 1000 before removing from view

The materialized view can be maintained without accessing the base table Part only during INSERT operations or when a cost is increased above 1000. For updates below 1000 and deletions, the base table must always be checked because one part can have multiple contracts with different prices.